

Date: 23.10.2025 (Thursday)

UNIVERSITY OF DHAKA

3rd Year B. S. (Honors) Incourse Examination 2025

Subject: CSE Course Title: Multivariable Calculus and Geometry

Course No: MATH 3105 Total marks: 25 Time: 1 Hour 30 minutes

Answer all the questions.

Q1.	a. Find parametric equations of the line L passing through the points $P(2,4,-1)$ and $Q(5,0,7)$. Also, find where does the line intersect the xy -plane? b. Find an equation of the plane through the points $P(1,2,-1)$, $Q(2,3,1)$, and $R(3,-1,2)$	3
Q2.	Identify the level curves of $f(x,y) = \sqrt{x^2 + y^2}$. Sketch a few of them.	2
Q3.	Define Curvature. What does curvature physically represent? Show that the curvature of a circular helix is constant.	4
Q4.	What is meant by the limit along a path? Provide the appropriate paths used to verify the existence of a limit for a function of two variables. Consider the function $f(x,y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$ Check the limit and continuity of the function $f(x,y)$ at the origin.	5
Q5.	a. Find the work done when a force $\vec{F} = \langle 3, 4, 0 \rangle$ moves an object through displacement $\vec{d} = \langle 2, 0, 1 \rangle$. b. Write the equation of a line through the origin and perpendicular to the plane $x + 2y + 3z = 6$. c. Find the angle between the planes $x + y + z = 0$ and $2x - y + 3z = 0$.	3
Q6.	Suppose that $w = x^2 + y^2 - z^2$ and $x = \rho \sin \beta \cos \gamma$, $y = \rho \sin \beta \sin \gamma$, $z = \rho \cos \beta$ Use appropriate forms of the chain rule to find w_ρ , w_β , and w_γ .	3
Q7.	i. If the projection of $\vec{a}$ on $\vec{b}$ is zero, what does that imply? ii. How does a vector-valued function differ from a scalar function? iii. How is the derivative of a vector function interpreted geometrically? iv. If curvature is zero, what kind of path does the particle follow? v. Explain the geometric meaning of a partial derivative.	5

Best of Luck